```python
#[EXERCISE 1]

from collections import Counter, defaultdict
import pandas as pd

test_sources = [f"{LAST_NAME}_T{i}" for i in range(1, 6)]
measurement_categories = ["TEMPERATURE", "VOLTAGE", "CURRENT", "PRESSURE", "SPEED"]

test_records = []
for source_idx, source in enumerate(test_sources):
    for category_idx, category in enumerate(measurement_categories):
        artist_value = artist_positions[(source_idx + category_idx) % len(artist_positions)]
        value = SEED_NUM + artist_value + (source_idx + 1) * (category_idx + 1)
        test_records.append({
            "source": source,
            "category": category,
            "value": value
        })

test_records.append(test_records[0].copy())

generated_df = pd.DataFrame(test_records)
print("GENERATED TEST DATA")
print(generated_df.to_string(index=False))

organized_df = generated_df.sort_values(["source", "category", "value"]).reset_index(drop=True)
print("\nORGANIZED TEST DATA")
print(organized_df.to_string(index=False))
```

Python

Activity4_Appendix.ipynb > %pip install pandas

Jupyter Notebook

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```
...   GENERATED TEST DATA
         source    category  value
      Ganade_T1 TEMPERATURE     12
      Ganade_T1     VOLTAGE     24
      Ganade_T1     CURRENT     12
      Ganade_T1    PRESSURE     22
      Ganade_T1       SPEED     29
      Ganade_T2 TEMPERATURE     24
      Ganade_T2     VOLTAGE     13
      Ganade_T2     CURRENT     24
      Ganade_T2    PRESSURE     32
      Ganade_T2       SPEED     30
      Ganade_T3 TEMPERATURE     12
      Ganade_T3     VOLTAGE     24
      Ganade_T3     CURRENT     33
      Ganade_T3    PRESSURE     32
      Ganade_T3       SPEED     24
      Ganade_T4 TEMPERATURE     22
      Ganade_T4     VOLTAGE     32
      Ganade_T4     CURRENT     32
      Ganade_T4    PRESSURE     25
      Ganade_T4       SPEED     42
      Ganade_T5 TEMPERATURE     29
      Ganade_T5     VOLTAGE     30
      Ganade_T5     CURRENT     24
      ...
      Ganade_T2     VOLTAGE                1          13        13.0
      Ganade_T3     VOLTAGE                1          24        24.0
      Ganade_T4     VOLTAGE                1          32        32.0
      Ganade_T5     VOLTAGE                1          30        30.0
```